SMART INDIA HACKATHON • IDEA PRESENTATION

# ORCA

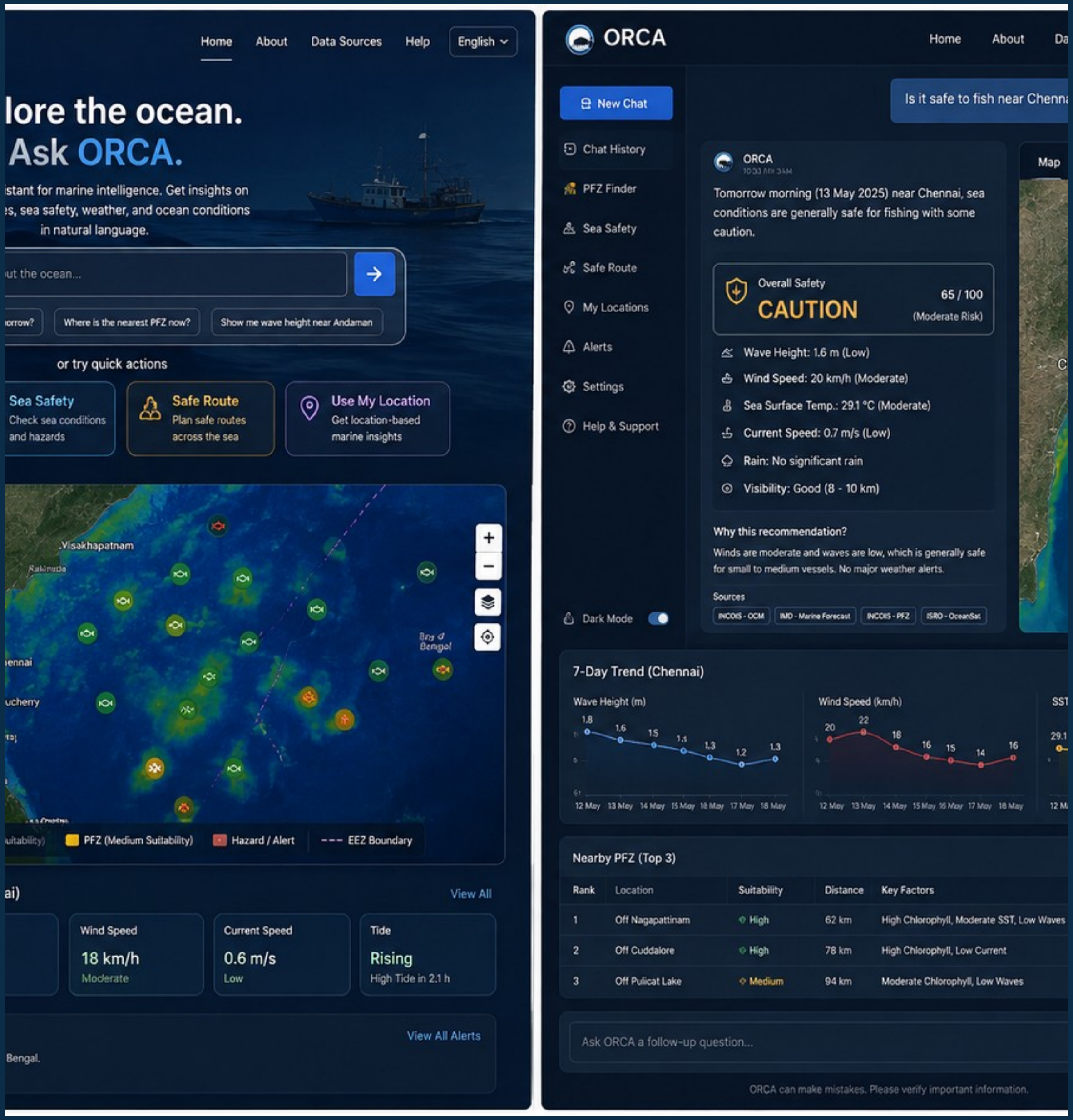
## Agentic AI for Ocean Intelligence

Ask about the ocean in simple language. ORCA connects the right information, analysis and maps to help you decide what to do next.

**SIH26176 • SOFTWARE • MISCELLANEOUS**

Problem: ORCA Marine EcOsystem Reasoning with Collaborative Agents

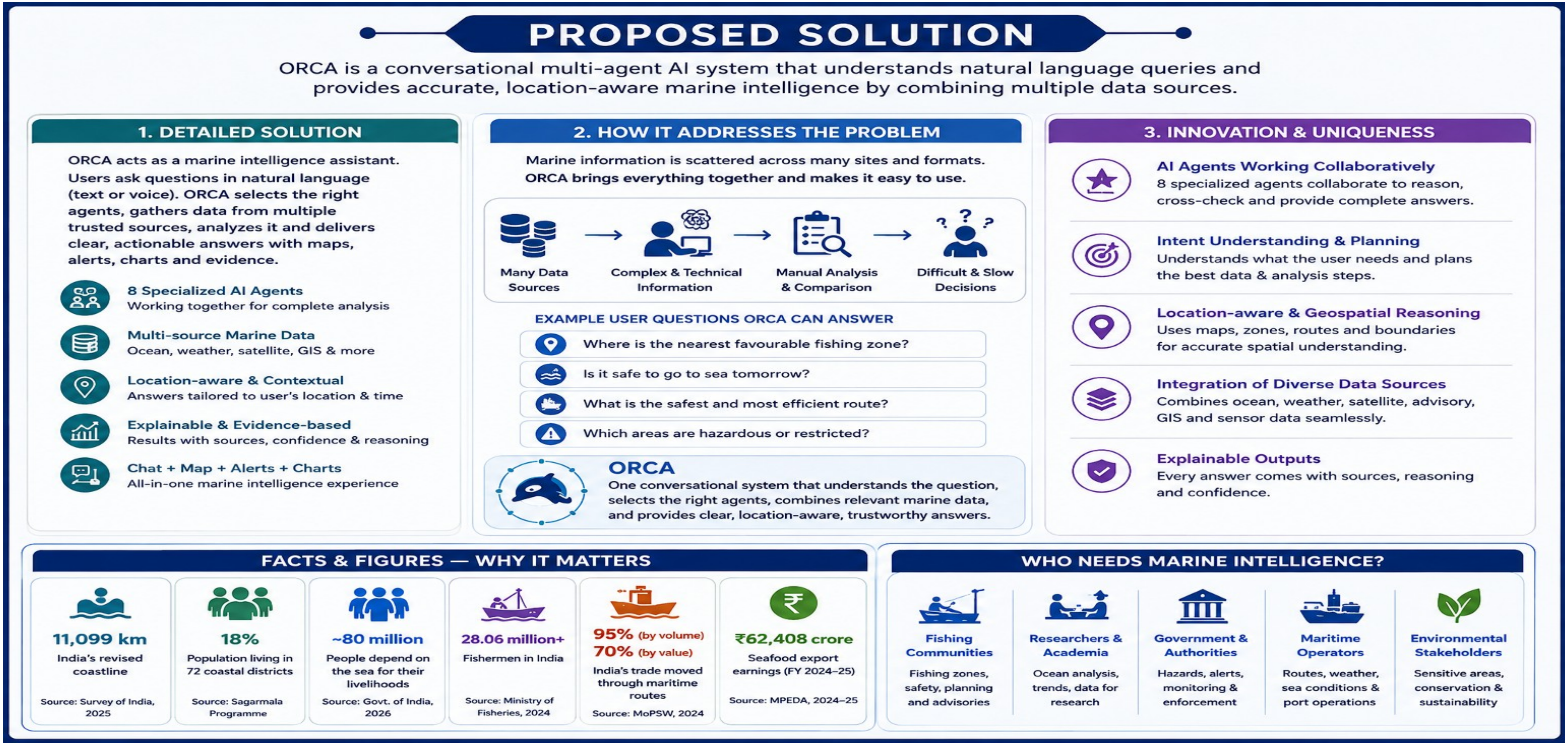
Team ID: ___________ Team Name: CodeCatalysts



SLIDE 2

# Proposed Solution

Exact generated solution visual — kept as the main design reference.

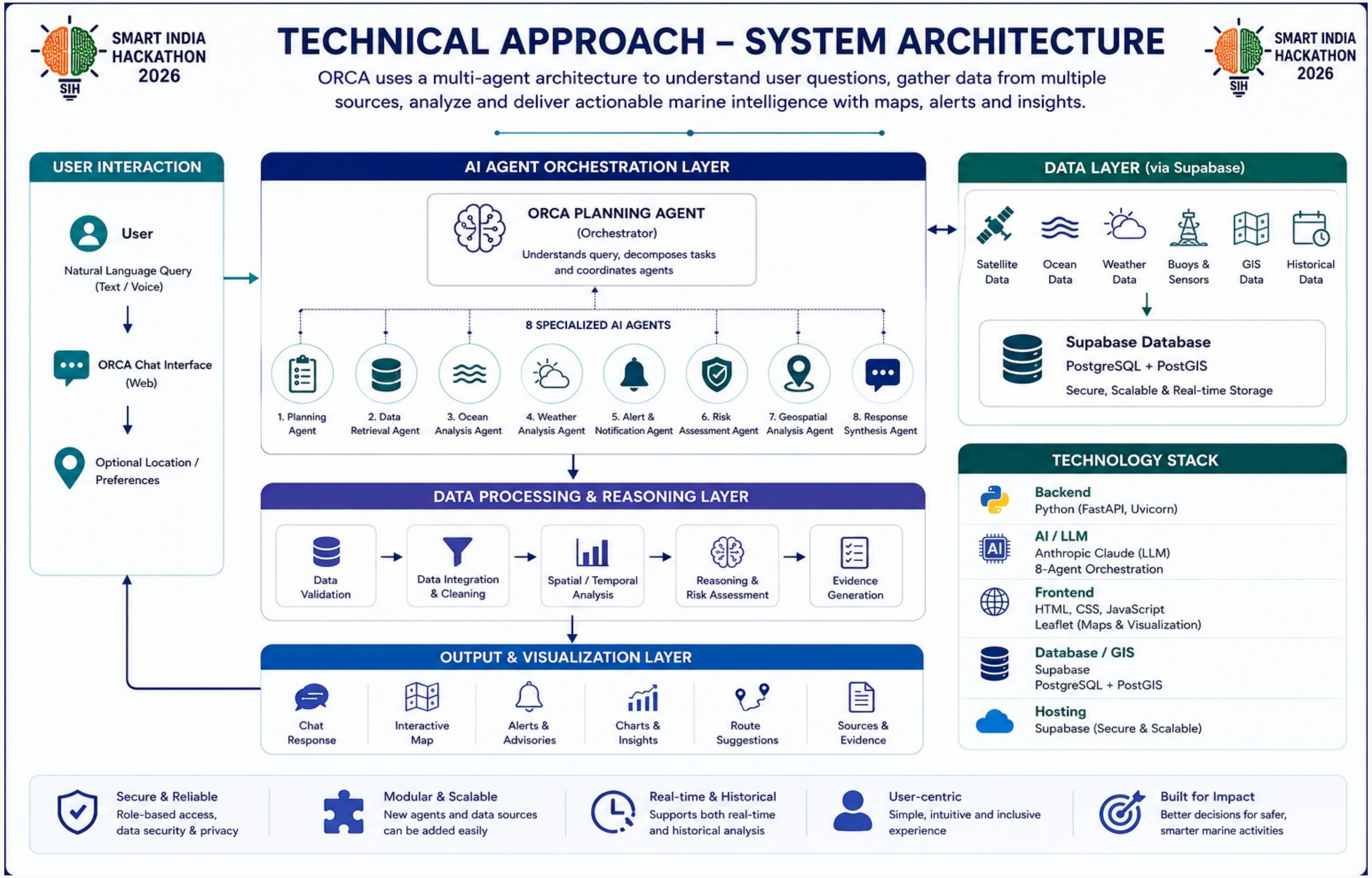


# Technical Approach

Editable implementation flow on the left; exact generated architecture visual on the right.

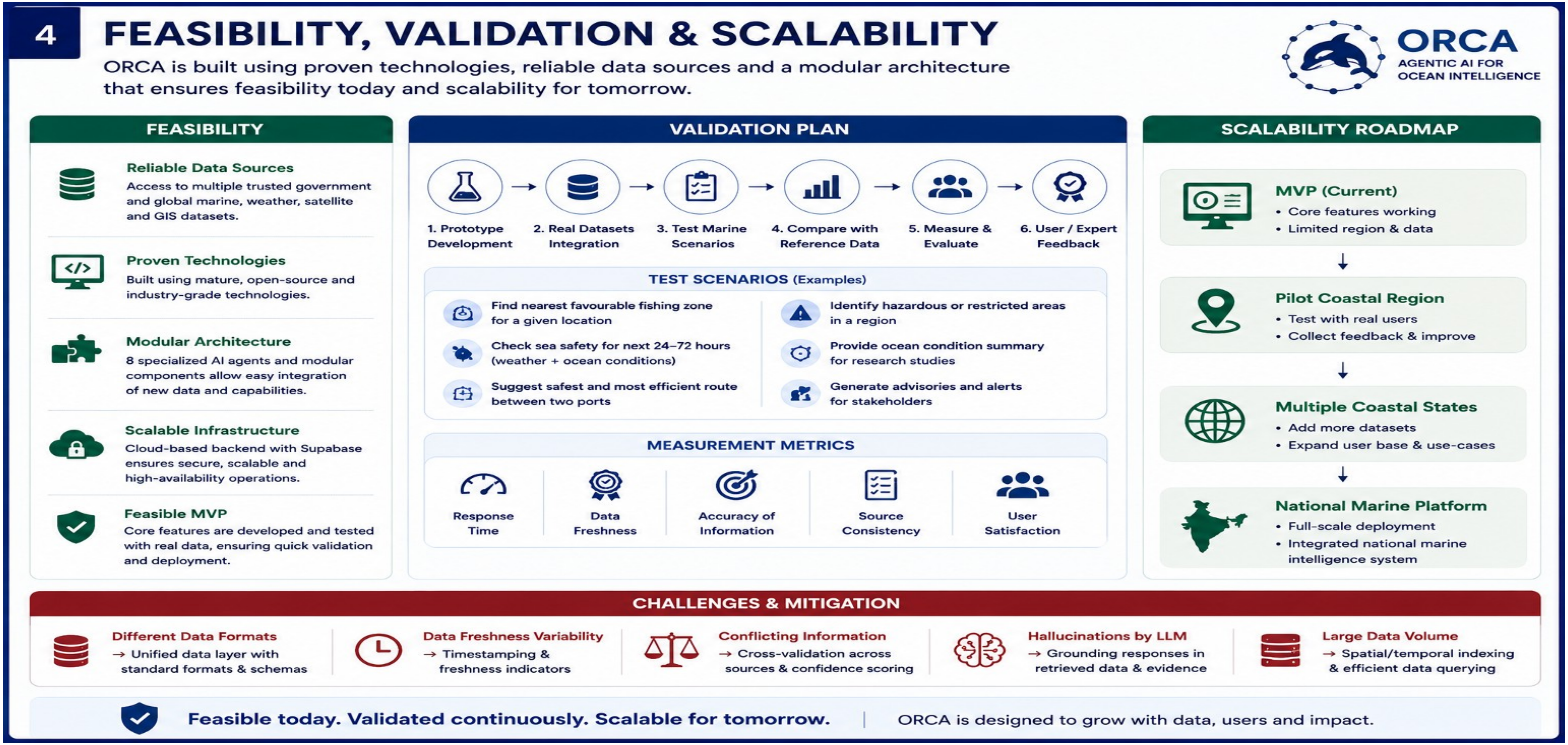
## IMPLEMENTATION FLOW

1. **USER INPUT**
   Question + optional location
2. **PLANNING AGENT**
   LLM understands the request + builds the task plan
3. **8 SPECIALISED AGENTS**
   Work together on the required task
4. **DATA**
   Marine + weather + satellite + GIS
5. **VALIDATE & ANALYSE**
   Check source, time, location + relationships
6. **REASON & ASSESS**
   Combine results, constraints and risk
7. **OUTPUT**
   Chat + map + alerts + route + evidence



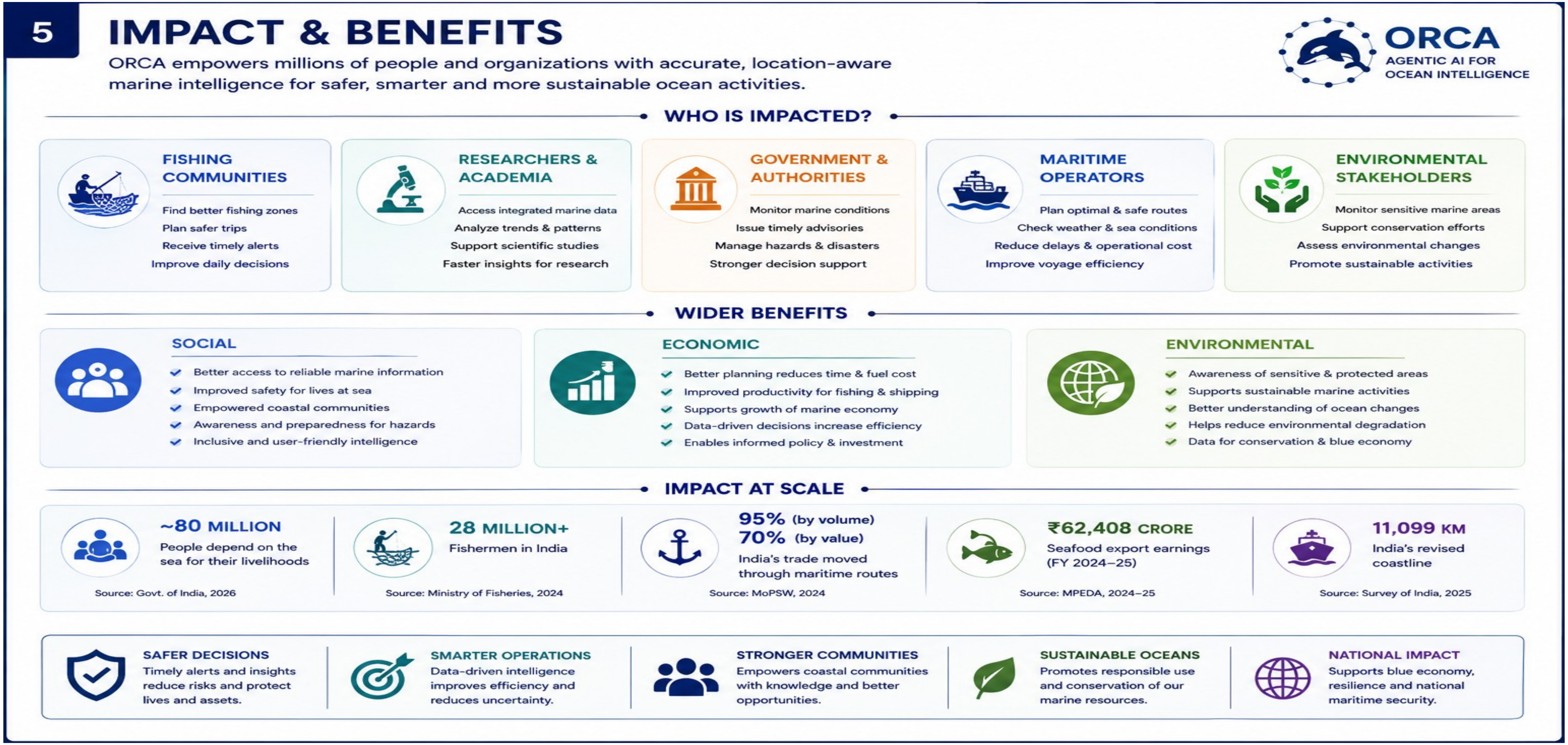
# Feasibility, Validation & Scalability

Exact generated feasibility visual — this is our working slide design.



# Impact & Benefits

Exact generated impact visual — with stakeholder groups, wider benefits and scale.



SLIDE 6

# Research & References

Exact generated research/reference visual — source names can be finalized later.